

Taiwan Semiconductor Manufacturing Company Limited

Semiconductor Manufacturing Report — Fiscal Year 2023

Year Ending December 31, 2023

1. Company Overview

Taiwan Semiconductor Manufacturing Company Limited (TSMC) is the world's largest dedicated independent semiconductor foundry, headquartered in Hsinchu, Taiwan. TSMC manufactures integrated circuits for customers who design their own chips but do not operate fabrication facilities. TSMC's customers include the world's leading fabless semiconductor companies and integrated device manufacturers.

TSMC operates on a pure-play foundry model — it does not design or sell chips under its own brand. Instead, TSMC manufactures chips to the exact specifications provided by its customers. TSMC's advanced process technology leadership, particularly its 3-nanometer (N3) and 2-nanometer (N2) nodes, makes it the preferred or sole-source manufacturer for the most advanced chips produced by Apple, Nvidia, AMD, Qualcomm, and other leading designers.

2. Key Customers and Revenue Concentration

Apple Inc. is TSMC's largest single customer, accounting for approximately 25% of TSMC's total revenue in fiscal year 2023. Apple uses TSMC exclusively to manufacture its Apple Silicon chips, including the A17 Pro used in iPhone 15 Pro and the M3 family used in Mac and iPad devices. The Apple-TSMC relationship is deeply interdependent — Apple commits to purchasing TSMC's most advanced process capacity years in advance, providing TSMC with the revenue certainty required to invest in next-generation fabrication equipment.

Nvidia Corporation is TSMC's second-largest customer by revenue contribution in fiscal year 2023, with TSMC manufacturing Nvidia's H100 and A100 data center GPU chips. Nvidia's explosive revenue growth driven by artificial intelligence compute demand has made it one of the fastest-growing customers in TSMC's history. Nvidia's H100 GPU, manufactured on TSMC's 4-nanometer process node, became the most sought-after semiconductor product globally in 2023 due to demand from hyperscale cloud providers and AI research institutions.

Customer	Estimated Revenue Share	Primary Products Manufactured	Process Node
Apple Inc.	~25%	A17 Pro, M3, M3 Pro, M3 Max	N3, N3E
Nvidia Corporation	~15%	H100, A100, L40S GPUs	N4, N4P

AMD	~10%	EPYC, Ryzen CPUs; RX GPUs	N5, N4
Qualcomm	~8%	Snapdragon 8 Gen 3	N4
Broadcom / Others	~42%	Various networking, RF chips	Various

3. Equipment and Technology Suppliers

TSMC's ability to manufacture chips at advanced process nodes depends on a small number of highly specialised equipment suppliers. The most critical dependency is ASML Holding N.V., headquartered in Veldhoven, Netherlands. ASML is the sole global manufacturer of extreme ultraviolet (EUV) lithography machines, which are essential for patterning circuit features smaller than 7 nanometers.

TSMC is the largest purchaser of ASML's EUV systems globally. Each EUV machine costs approximately USD 150 million to USD 200 million and weighs approximately 180 metric tonnes. A single TSMC fab requires between 30 and 50 EUV machines to operate at full production capacity. TSMC's capital expenditure for fiscal year 2023 was approximately USD 32.1 billion, with EUV equipment procurement representing a significant portion of this figure.

Other critical equipment suppliers to TSMC include Applied Materials, Lam Research, and KLA Corporation, all headquartered in the United States. These companies supply deposition, etch, and inspection equipment respectively. Export control restrictions imposed by the United States government on semiconductor equipment exports to China affect TSMC's competitors but also create dependency risks if US-China trade relations deteriorate further.

4. Geographic Concentration and Geopolitical Risk

TSMC's primary manufacturing operations are concentrated in Taiwan, with major fabs in Hsinchu, Taichung, and Tainan. Taiwan's geopolitical status — claimed by the People's Republic of China as a province — creates a unique and material risk for TSMC and for the global semiconductor supply chain. A military conflict or naval blockade of Taiwan would disrupt or halt TSMC's manufacturing operations, affecting the supply of advanced chips to Apple, Nvidia, AMD, Qualcomm, and hundreds of other customers worldwide.

To reduce geographic concentration, TSMC is building new fabrication facilities in the United States (Phoenix, Arizona) and Japan (Kumamoto, Kyushu) with financial support from the respective governments. The Arizona Fab 1 is targeted to begin N4-process production in 2025, and Fab 2 in Arizona is planned for N2-process production in 2028. However, these international facilities will represent a small fraction of TSMC's total manufacturing capacity for the foreseeable future.

5. Financial Summary

Metric	FY2023	FY2022	Change
Net Revenue	USD 69.3B	USD 75.9B	-8.7%
Gross Profit	USD 32.3B	USD 38.6B	-16.3%
Gross Margin	46.6%	50.9%	-4.3 pp
Net Income	USD 26.5B	USD 35.2B	-24.7%
Capital Expenditure	USD 32.1B	USD 36.3B	-11.6%
R&D Expenditure	USD 5.9B	USD 5.4B	+9.3%

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